Course Code	Name of the Course			
216U03C304	Microprocessor and Microcontroller			
Teaching Scheme	TH	P	TUT	Total
	03	0	0	03
Credits Assigned	03	0	0	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		20	30	50	100

Course pre-requisites:

Fundamentals of Digital Electronics, Binary and Hexadecimal number systems.

Course Objectives:

Microprocessors and microcontrollers are the integral part of any embedded systems. The course will enable students to learn basic programming of Microprocessor, Microcontroller and essential fundamentals for interfacing different peripheral devices to develop applications.

Course Outcomes (CO):

After successful completion of the course the student will be able to:

CO1. Understand architecture of 8086
CO2. Understand architecture and memory organization of 8051
CO3. Write assembly language programs for 8086 and 8051
CO4. Understand Timer operations ,serial communication and their programming
CO5. Interface peripherals like ADC, DAC, Sensors etc. with microcontroller and develop various applications

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1		**Microprocessor 8086 architecture**	10	CO1
	1.1	Introduction to microprocessors and microcontrollers , evolution of microprocessors		
	1.2	Architecture, organization, Memory segmentation, pin configuration of 8086 , Memory banking		
	1.3	Minimum mode and Maximum mode of 8086		
	1.4	Read and Write bus cycle		
	1.5	Interrupt structure of 8086		
2		**Instruction set and programming of 8086**	10	CO3
	2.1	Opcode, operands, instruction word bytes, instruction cycle, fetch operation, execute operation and machine cycle		
	2.2	Addressing modes: Register, direct, register indirect, immediate, based, indexed, based indexed, based indexed and displacement		
	2.3	Instruction Set of 8086: Classification and description of instructions		
	2.4	Assembly Language programming for 8086		
3		**8051 Microcontroller**	10	CO2
	3.1	Overview, features and memory organization of 8051		
	3.2	Architecture, Pin diagram, addressing modes and assembler directives		
	3.3	Instruction set of 8051: Arithmetic, logical data transfer, branching and Boolean		
	3.4	Assembly language programming for 8051		
`		#Self-learning topic: Programs using subroutine		
4		**Timers, Serial Communication and Interrupts in 8051**	10	CO4
	4.1	Timers in 8051, timer modes and programming in assembly language		
	4.2	Basics of serial communication, 8051 connection to RS232, serial data transfer mode and serial port programming in assembly language		
	4.3	Interrupt structure in 8051, programming timer, external hardware and serial communication interrupts		
5		**8051 peripheral interfacing**	05	CO5
	5.1	Interfacing of ADC-0808 with 8051		
	5.2	Interfacing of DAC-0808 with 8051		
	5.3	Interfacing of 14-pin LCD with 8051		
	5.4	Interfacing of 4X4 matrix Keyboard with 8051		
		#Self-learning topic: Interfacing of Bluetooth module and different Sensors		
		Total	45	

 # Self Learning: Students should prepare all self-learning topics on their own. Self learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and or laboratory experiments

Reference Books

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1	Kenneth J. Ayala	*The 8086 Microprocessor: Programming & Interfacing The PC*	Cengage Learning, India	3 rd Edition, 2008
2	Douglas V Hall	*Microprocessors and Interfacing*	Tata McGraw Hill Education, India	2 nd Edition, 2007
3	Muhammad Ali Mazidi Janice Gillispie Mazidi Rolin D. McKinlay	*The 8051 Microcontroller and Embedded Systems*	Pearson, India	2 nd Edition, 2008
4	Kenneth J. Ayala	*The 8051 Microcontroller*	Cengage Learning,India	3 rd Edition, 2004